

# TAL KLEIN

## Data Scientist & Machine Learning Engineer

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### SUMMARY

ML researcher and data scientist specializing in GenAI, LLMs, and deep learning. Built production-scale NLP pipelines with PyTorch and GPT-based inference across 6,000+ documents. Hands-on experience with RAG architectures, LoRA/QLoRA fine-tuning, prompt engineering, and agentic systems (LangGraph). Strong foundation in Python, scikit-learn, TensorFlow, and model optimization.

### EDUCATION

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| <b>Ben-Gurion University of the Negev</b><br><i>M.Sc. Software &amp; Information Systems Engineering (Specialization: Data Science &amp; NLP) — <b>Meitar Excellence Program</b></i><br>• <b>Thesis:</b> LLM-driven feature extraction and causal inference for sentencing analysis in Hebrew court verdicts | 2024 – 2026<br>GPA: 92 |
| <b>Ben-Gurion University of the Negev</b><br><i>B.Sc. Data Engineering</i><br>• <b>Key courses:</b> Machine Learning, Deep Learning, NLP, Computer Vision, Algorithms, Statistics, Big Data                                                                                                                  | 2021 – 2025<br>GPA: 86 |

### PROFESSIONAL EXPERIENCE

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| <b>Data Scientist &amp; ML Researcher — NLP &amp; Causal Inference</b><br><i>Socially Embedded Lab, Ben-Gurion University</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2024 – Present<br>Beer Sheva, Israel |
| <ul style="list-style-type: none"><li>Built an end-to-end NLP pipeline extracting 30+ structured features from 4,000+ Hebrew court verdicts via GPT-5-mini batch inference, achieving <b>0.92 F1</b> on a 125-document annotated test set</li><li>Refined extraction prompts through error analysis on a held-out dev set, improving accuracy by <b>11 points</b> over a zero-shot baseline</li><li>Ran causal inference via propensity score matching with sensitivity analysis (Rosenbaum bounds, E-values) and MICE imputation to quantify sentencing disparities</li></ul> |                                      |

### PROJECTS

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| <b>LoRA/QLoRA Fine-Tuning of Gemma-3 — Continued Pre-training &amp; SFT</b>   <i>Python, PyTorch, PEFT/LoRA, TRL, Hugging Face</i>   <a href="#">GitHub</a> <ul style="list-style-type: none"><li>Domain-adapted Gemma-3 to the Sherlock Holmes canon via LoRA continued pre-training, with closed-form GPU memory analysis across model sizes from 1B to 13B</li><li>Built an SFT + retrieval-grounded QA pipeline raising closed-book exact-match from <b>0.009 to 0.611</b>; ran 15+ ablations (rank, target modules, data scale, catastrophic forgetting)</li></ul>                                                           |
| <b>Wikipedia-Scale Information Retrieval Engine</b>   <i>Python, BM25, TF-IDF, PageRank, Word2Vec, GCP</i>   <a href="#">GitHub</a> <ul style="list-style-type: none"><li>Implemented and deployed a hybrid search engine over <b>6.4M Wikipedia documents</b> on GCP, combining BM25, TF-IDF, PageRank, and PageViews into a weighted final ranking</li><li>Optimized retrieval via multi-threading and heap-based ranking, cutting query latency <b>from 9.8s to 1.24s (8x)</b> while maintaining <b>Precision@10 of 0.63</b></li></ul>                                                                                         |
| <b>The Sound of Silence — LLMs vs. Human Silence Annotators Research</b>   <i>Python, LLMs, Hebrew NLP, AssemblyAI</i> <ul style="list-style-type: none"><li>Built an end-to-end Hebrew-dialogue NLP pipeline (audio diarization, dynamic silence detection, custom web annotation platform) and released the first annotated silence dataset (315 segments, 1,425 labels) on Hugging Face</li><li>Benchmarked Gemini-2.0-flash on 6-class silence classification (<b>Cohen's kappa = 0.13</b>) against untrained human annotators (<b>Cohen's kappa = 0.26</b>), showing a measurable LLM gap in non-verbal pragmatics</li></ul> |
| <b>Next Frame Video Prediction</b>   <i>Python, BLIP2, YOLO, Computer Vision</i>   <a href="#">GitHub</a> <ul style="list-style-type: none"><li>Built a multimodal video prediction model combining YOLO object detection with BLIP2 captioning, fusing visual and textual embeddings to predict narrative continuity across frames</li></ul>                                                                                                                                                                                                                                                                                     |

### TECHNICAL SKILLS

**Languages & Tools:** Python (advanced), R, SQL, Bash, Git, Docker, GCP, PostgreSQL  
**ML & DL:** PyTorch, scikit-learn, XGBoost, Hugging Face Transformers, TensorFlow  
**LLMs & GenAI:** LLM APIs, prompt engineering, structured outputs, RAG, LoRA/QLoRA fine-tuning, LangGraph, LLM evaluation  
**NLP:** Hebrew NLP, Transformers, Hugging Face, embeddings  
**Computer Vision:** OpenCV, YOLO, BLIP-2, CLIP  
**Data & Stats:** Pandas, NumPy, Selenium, ETL, MICE imputation, propensity score matching, Bayesian methods, A/B testing  
**Spoken:** Hebrew (native), English (fluent)

### ADDITIONAL EXPERIENCE

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| <b>IT Support &amp; Help Desk</b><br><i>Matrix</i> <ul style="list-style-type: none"><li>Monitored fault patterns and improved incident response across organizational systems</li></ul>                                      | 2020 – 2021<br>Israel |
| <b>Communications &amp; Operations NCO</b><br><i>Israel Defense Forces (IDF)</i> <ul style="list-style-type: none"><li>Led system upgrades and supervised a team of 8 soldiers managing data and network operations</li></ul> | 2017 – 2020<br>Israel |

### VOLUNTEER

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| <b>Volunteer Puppy Raiser (Foster Family Program)</b><br><i>Israel Guide Dog Center for the Blind</i> | 2025 – Present |
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